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Review article

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Three-photon microscopy: an emerging technique for deep intravital brain imaging

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Abstract

Understanding brain function and pathology requires observation of cellular dynamics within intact neural circuits. Although two-photon microscopy revolutionized mammalian in vivo brain imaging, its limitation to upper cortical layers has restricted access to many important brain regions. Three-photon microscopy overcomes this constraint, enabling minimally invasive yet high-resolution visualization of the deep cortical and subcortical structures that are crucial for higher-order brain functions. This emerging technology opens new avenues for investigating fundamental aspects of neuroscience, from circuit dynamics to disease mechanisms. Here, we examine how three-photon microscopy has started to transform our ability to investigate neural circuits, glial biology, and oncological and neuroimmune interactions in previously inaccessible brain regions, primarily in the mouse, but also in other model organisms. We discuss current technical challenges, recent innovations and future applications that promise to bring us greater understanding of the living brain.

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Introduction

The evolution of optical microscopy has been fundamental to our understanding of brain function, providing powerful ways to visualize neural dynamics with cellular and subcellular resolution across scales[1]. Optical methods are uniquely suited for studying neurons and their fine substructures, making them indispensable for investigating how brain cells function in health and disease (Box 1). In vivo optical imaging has further transformed neuroscience by enabling the direct observation of cellular processes in living brain tissue with exceptional spatial and temporal resolution. By combining fluorescent reporters with imaging, researchers can measure cellular activity and network dynamics in the intact brain[2–5], track the same cells and structures over extended periods[6–10], and study neural plasticity, learning[11] and degeneration. Moreover, the combination of in vivo optical imaging with genetic tools offers the possibility to selectively visualize and investigate genetically defined cell types[12,13]. Together, these approaches have provided crucial insights into how neural circuits adapt and respond to various conditions[14] and stimuli, establishing optical microscopy as a key technique in modern neuroscience research.

Imaging deep into live biological tissue presents a substantial technical challenge owing to the strong light scattering caused by heterogeneous tissue components. Historically, this has limited high-resolution optical modalities such as confocal microscopy to thin tissue sections or superficial cortical layers[15]. The advent of multiphoton microscopy (MPM), and in particular two-photon microscopy[16] (2PM), has greatly expanded the depth at which high-resolution optical imaging can be achieved, particularly for in vivo applications[17]. By using longer wavelengths of light than traditional single-photon-based microscopy modalities, MPM minimizes scattering in biological tissues, allowing deeper penetration into the brain. This makes it possible to image neural structures that are located several hundred micrometres below the surface of the brain (such as cortical layers I–IV in the mouse[18]) with subcellular resolution, which would be difficult to achieve using single-photon microscopy. Moreover, when combined with genetically encoded calcium indicators[19] or voltage-sensitive reporters[20], MPM can be used to visualize neuronal activity in real-time with remarkable sensitivity. For example, MPM can allow calcium and voltage dynamics, including individual action potentials[21,22], to be observed deep within the brain. This has opened new avenues for studying the dynamics of neural circuits during behaviour and, more generally, across cell types, development and disease.

Despite its advantages, 2PM imaging is, in practice, limited to upper cortical layers, beyond which strong out-of-focus background fluorescence lowers the signal-to-background ratio. As a result, to image in deep brain regions, researchers typically remove the cortex[23] and implant a cannula or a miniature-sized gradient index lens[24], possibly altering the brain environment. Therefore, in vivo optical imaging of deep-seated mammalian brain regions, such as the hippocampus, is still a major technical challenge. Here, the relatively recent emergence of three-photon microscopy (3PM) presents a transformative approach[25,26]. Three-photon microscopy relies on a higher-order multiphoton excitation, in which three photons are absorbed simultaneously to excite a fluorophore or generate a third-order harmonic signal contrast[27,28]. This higher nonlinear light–matter interaction can suppress detrimental out-of-focus background signals, thereby enhancing the signal-to-background ratio in highly scattering tissues and enabling deeper tissue penetration (Box 2).

This Review will introduce 3PM and its underlying key physical principles, discuss the recent emergence and establishment of this optical deep brain imaging technique, and provide an overview of its advantages and opportunities, with a focus on its applications for studying neuroscience in its various dimensions. In particular, we will highlight how this technology and the first studies using this approach have started to provide new insights into the function, organization and pathology of the brain. We will also showcase the novel applications and technological breakthroughs that 3PM could catalyse that may further elucidate the structural and functional principles of the nervous system.

Principles and technical advances of 3PM

The main obstacles for deep brain imaging are the scattering and absorption of ballistic photons by biological tissue, which attenuate the amount of excitation light reaching the focus and, therefore, the amount of signal that can be generated. Here, the so-called attenuation coefficient effectively quantifies the overall loss of light when passing through tissue[29,30]. 3PM supports greater imaging depth than 2PM for several reasons. First, 3PM uses longer excitation wavelengths, which generally lowers scattering within heterogeneous biological tissue (Fig. 1a). Second, the higher-order nonlinear excitation of 3PM markedly reduces the out-of-focus background, which is otherwise prominent when optical power is increased at the surface to maintain excitation energy at depth. This increase in optical power is necessary because the amount of ballistic light available to excite a fluorophore decreases exponentially with depth ($\sim\mathrm{Exp}(-z/l_e)$, where z denotes the imaging depth and l_e the characteristic attenuation length of the tissue). The out-of-focus background is generated by the cone of excitation light below and particularly above the focus. The requirement for three instead of two photons to raise the energy of the fluorophore makes fluorescence excitation out-of-focus less probable for three-photon excitation (3PE) than for three-photon excitation (2PE) (Fig. 1b). Four-photon (4P) excitation can, in principle, provide even stronger out-of-focus background rejection and, therefore, has proven useful in strongly scattering tissues such as skin or bone[31]. However, the 4P absorption cross section is usually much lower than that of 3PE, yielding fluorescence signals that are prohibitively low for most practical applications.

For effective 3PE of most widely used green and red fluorescent proteins, excitation wavelengths within two spectral windows (at approximately 1,300 nm and 1,700 nm, respectively) are typically used[32]. Both windows avoid the strong absorption in brain tissue (primarily owing to water) that occurs around 1,450 nm (Fig. 1a). The excitation wavelength has a much larger impact on imaging depth than the wavelength of the generated signal or fluorescence emission[33]. In addition, 3PE at 1,700 nm has the advantage of reduced scattering, when compared to 3PE at 1,300 nm. Although the overall 3PE cross section[34], and thus the signal generation efficiency of 3PM, is low, careful optimization of excitation light parameters can yield a fluorescence signal that is sufficient for practical imaging even at depth (>1 mm), while maintaining excitation levels at the surface that are low enough to avoid deleterious heating, photodamage and phototoxicity[35–37]. These parameters are discussed in more detail in Box 2, and a comprehensive overview of fluorophores and reporters that have been successfully used in 3PM is provided in Supplementary Table 1.

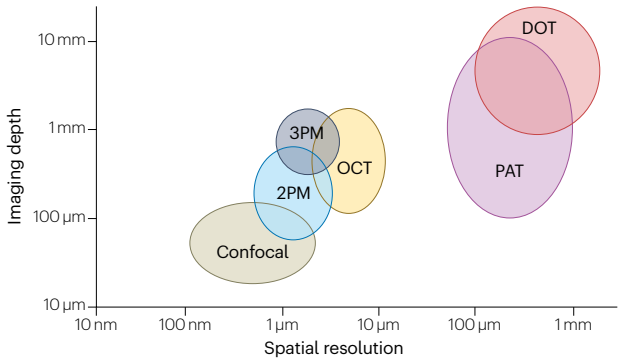
The first demonstration of the concept of 3PM in biological tissues dates back to the mid-1990s[38–40]; however, it was not until 2013 that the technique became mature enough for in vivo brain imaging[25,41], an advance that was enabled by the development of a

Box 1 | Deep brain imaging methods across scales

Various approaches to examine mammalian brain tissue exist. In comparison with non-optical imaging methods, such as magnetic resonance imaging, ultrasound and X-rays, optical methods offer very high (typically subcellular) spatial resolution, high sensitivity and — in the case of fluorescence — endogenous or exogenous contrasts that allow the visualization of various structures across spatial scales. Optical methods also allow brain tissue to be studied in vivo non-invasively and over prolonged time periods. Below, we briefly describe the main optical technologies capable of 'deep' brain imaging (beyond 500 µm depth). Their performance is summarized in the figure, wherein the lowest spatial resolution of each technique is illustrated and the brain features that can be visualized are listed. Note that we have also included confocal microscopy here that, although limited to very superficial layers, serves as reference technique.

Nonlinear optical microscopy, also known as multiphoton microscopy (MPM), relies on nonlinear light–matter interactions involving two or more photons. Owing to the intrinsic properties of these interactions, the signals are confined to a highly localized region around the focus[156]. Together with the reduced scattering enabled by long excitation wavelengths, this makes MPM the gold standard for in vivo brain imaging. Three-photon microscopy (3PM) enables greater imaging depths than two-photon microscopy (2PM), giving access to subcortical brain structures, and can be used to visualize the cellular microenvironment. The combined energy of the excitation photons excites fluorophores (Supplementary Table 1) that can be exogenous (for example, fluorescent proteins expressed by genetically modified organisms) or endogenous (such as the naturally occurring autofluorescence of cells and tissues containing NADH, FAD or other molecules). The photons can also generate label-free signals that depend on the sample properties. For instance, second harmonic generation occurs in non-centrosymmetric, highly organized tissues such as collagen fibres, whereas third harmonic generation occurs at the interfaces of optically distinct media. These label-free signals can give valuable anatomical context and complement fluorescence imaging; however, their specificity and interpretation can be complex and non-trivial.

Optical coherence tomography (OCT)[157] provides detailed structural images of biological tissues at near-microscopic resolution, with the ability to image at depths approaching 1 mm in the mouse. It uses low-coherence interferometry to detect backscattered light from different tissue depths, which can be used to form 3D images.



	Dendritic spine	Neuronal process	Neuron	Full cortical column	Brain region
Confocal	✓	✓	✓	✗	✗
2PM	✓	✓	✓	✗	✗
3PM	✓	✓	✓	✓	✗
OCT	✗	✓	✓	✓	✗
PAT	✗	✗	✗	✗	✓
DOT	✗	✗	✗	✗	✓

However, its label-free signal typically lacks strong contrast and/or molecular specificity in mammalian brain tissue.

Diffuse optical tomography (DOT)[158,159] measures the propagation of near-infrared light through tissues and can be used to visualize brain activity non-invasively. In DOT, light is directed into the brain and an array of detectors measures the light that is absorbed or scattered as it passes through tissues. Such data provides insights into brain function, including changes in blood flow and oxygenation associated with neuronal activity. However, the spatial resolution of DOT is generally much lower than that of MPM, making it best suited for imaging large-scale functional changes rather than fine structural details.

Photoacoustic tomography (PAT)[160,161] combines optical excitation and ultrasound detection to provide mesoscopic-resolution images of tissues. In PAT, short pulses of laser light are absorbed by tissue components, causing localized heating and generating ultrasonic waves that are detected and used to reconstruct images of the tissue. For mouse brain imaging, PAT is particularly useful for visualizing the vasculature and monitoring changes in blood oxygenation[162]. It generally offers deeper tissue penetration than purely optical methods, but its spatial resolution is much lower, and techniques to label specific cells or circuits are limited.

tailored femtosecond laser source with sufficient pulse energies at the near-infrared wavelengths that are important for 3PM. The subsequent rapid development of 3PM for various bioimaging applications has been facilitated by the rise of commercially available energetic femtosecond laser sources with properties optimized for 3PE. Over the past decade, there has been remarkable technical advancement in 3PM methods and approaches, increasing the toolbox available to address questions in neuroscience. Systematic studies with respect to living-tissue optical properties[32,42] and safety limits[36] have established the preferred 3PE spectral windows, physiological tissue-damage thresholds, and optimal laser source parameters for in vivo 3PM[26] (Box 2).

Hardware improvements

The rise of near-infrared-suitable objectives, as well as scan lenses, tube lenses and gallium arsenide phosphide (GaAsP) photomultiplier tubes, has contributed to more efficient microscope designs and realizations[43]. Improvements in laser sources have been particularly instrumental and hold much potential for further development. For example, the ability to adapt the timing and energy of excitation laser pulses to modulate 'pulse trains' in real time has enabled researchers to tailor them to specific region(s) or cells. Thus, all the permissible laser power can effectively be concentrated on structures of interest[44]. This strategy can substantially reduce the overall laser power impinging on the brain, while increasing the imaging speed, field of view and number of neurons sampled.

Box 2 | The underlying physical principles of three-photon microscopy

The efficiency of signal generation in multiphoton microscopy depends on the multiphoton absorption cross section of the target fluorescence molecule[40,156]. Owing to the nonlinear multiphoton absorption process, the fluorescence excitation probabilities are extremely low. However, this can be overcome by concentrating the excitation in space and time. High spatial densities are generated by focusing a laser through a high-numerical aperture (NA) objective, whereas ultrashort pulsed lasers effectively concentrate light in the time domain, yielding pulse energies of a few nanojoules in less than 100 fs on the sample plane. In general, for samples that are within the 'thick sample limit' — that is, samples that are larger than the point spread function of the microscope — one can relate the time-averaged fluorescence photon flux $\langle F^{(n)}(t)\rangle$ to the incident power $P(t)$ under two-photon, three-photon and four-photon excitation, where n is the number of photons absorbed, as follows[34]:

$$\langle F^{(n)}(t)\rangle \sim \frac{\eta\sigma_n}{(f\tau)^{n-1}}\frac{(\mathrm{NA})^{2n-4}\langle P(t)\rangle^n}{\lambda^{2n-3}},$$

where η is the overall quantum efficiency of the process, σ_n the n-photon absorption cross section, λ the excitation wavelength, and f and τ the repetition rate and pulse duration of the laser, respectively. For two-photon excitation (2PE) of fluorescence:

$$\langle F^{(2)}(t)\rangle \sim \frac{\eta\sigma_2}{f\tau}\frac{\langle P(t)\rangle^2}{\lambda}.$$

We note that, for 2PE, there is no dependence on NA and that the excitation laser source parameters (spectral and temporal) have low influence. However, for three-photon excitation (3PE) of fluorescence, the signal becomes:

$$\langle F^{(3)}(t)\rangle \sim \frac{\eta\sigma_3}{(f\tau)^2}\frac{(\mathrm{NA})^2\langle P(t)\rangle^3}{\lambda^3}.$$

Besides the cubic versus quadratic dependence on the excitation power, several important differences can be observed in the comparison of the two modalities: these pertain mostly to NA, f and τ.

Unlike 2PE, the 3PE signal depends quadratically on NA, highlighting the importance of high-NA objectives to maximize excitation. This can be assured by overfilling the back aperture of the objective, but it must be balanced with the resulting power loss, which influences the maximum achievable imaging depth[15,163]. Moreover, care must be devoted to the temporal laser parameters in 3PE, as the signal depends quadratically on the inverse of the repetition rate and pulse duration. Typical pulse durations on the order of ~50–70 fs in the sample plane are required, and group delay dispersion management and compensation, therefore, become extremely important. The lower repetition rates used during 3PE, which are on the order of 0.1–2.0 MHz, compensate for the higher pulse energies at the focus (typically ~1 nJ), compared to 2PM (~0.1 nJ)[26]. However, there are practical limits to the maximum power (and pulse energy) that can be used, given the need to avoid heating, photodamage and phototoxicity that could affect the physiology of brain tissue[35,36]. Therefore, average laser powers at the surface should not exceed ~100 mW at 1,300 nm (or ~50 mW at 1,700 nm) for the mouse brain[37] to avoid heating-related photodamage. Furthermore, because focal pulse energies greater than 2–5 nJ have shown to induce photodamage[36], this parameter also needs to be diligently optimized and monitored during in vivo imaging. The photodamage and phototoxicity limits are highly sample- and experiment-dependent and, therefore, must be established for every specific case.

Adaptive optics

Another key technique in 3PM is the use of adaptive optics[45], which can correct wavefront distortions in the excitation light to restore (diffraction-limited) spatial resolution and increase the fluorescence signal and contrast at depth. Adaptation and optimization of this approach has yielded an unprecedented imaging depth of 1,450 µm (ref. 46) and the development of compact adaptive optics modules[47] for straightforward integration into different models of microscopes. Advanced wavefront sensing approaches[48,49] have the potential to further improve microscope resolution at depth and to correct a high number of aberration modes, which occur owing to the highly scattering nature of the brain. Typically, adaptive optics techniques can enhance fluorescence signal levels by twofold to fivefold and similarly improve the axial resolution of a microscope up to fivefold, especially in deep brain layers (Supplementary Fig. 1).

Acquisition speed

One prominent limitation of 3PM is imaging speed, mostly owing to the requirement for laser sources that deliver pulses with lower repetition rates (to remain below safe laser power thresholds) than those used for 2PM (Box 2). Faster volumetric imaging can be achieved by utilizing an axially elongated Bessel focus[50,51], allowing entire 3D volumes to be captured during a single 2D lateral scan and, thus, boosting imaging speed. However, this technique is applicable only to sparsely labelled samples as it lacks optical sectioning capability. Alternatively, remote focusing and/or dual-plane imaging systems[52,53] can enable faster volumetric or axial acquisitions while maintaining high optical resolution. When applied to neuronal (Ca^{2+}) recordings, careful modulation of the extent of the 3P focus (that is, the point-spread function) by underfilling the back aperture of the microscope objective (and, thus, reducing its effective NA), together with non-diffraction-limited spatial sampling, can optimize the trade-off between spatial resolution and imaging speed. This approach has enabled the population activity of ~350 neurons in the CA1 region of the mouse hippocampus to be captured at 17 Hz (ref. 54).

Fluorescent probes and image processing

Apart from these hardware-centric developments, recent work has reported an approach to enhance the resonance of 3PE. This allows simultaneous, and efficient, excitation of both green and selected red fluorescent molecules within the 1,300-nm spectral window[55], therefore removing the need to change the excitation wavelength between acquisitions for dual-colour imaging. Another advance has been provided by organic aggregation-induced emission nanoparticles, which have

been shown to have a high absorption cross section under 3PE[56], leading to high fluorescence signal levels. An overview of the functional reporters currently being used in 3PM can be found in Supplementary Table 1. Lastly, custom artificial intelligence (AI)-based image restoration (denoising) and classification approaches that are specifically tailored to the noise characteristics and label-free signals present in 3PM data have also been established[57], which will help to improve image signal-to-noise ratios (SNRs).

Applications: from circuits to disease

Since its pioneering application to in vivo mouse brain imaging in 2013 (ref. 25), 3PM has come a long way. Over the past decade, 3PM has reached a level of maturity that now makes it accessible to researchers beyond specialized technology development laboratories, enabling applications across fields and model organisms (Fig. 2; Supplementary Fig. 2).

The transformative potential of 3PM for neuroscience stems from its unique ability to non-invasively access deep brain structures that were previously observable only through highly invasive approaches. This capability changes the types of question that neuroscientists can address. Although 2PM revolutionized our understanding of superficial cortical circuits, many important neural processes involve deep cortical layers, white matter and subcortical structures, which together regulate higher cognitive functions, memory formation and emotional responses[58,59]. By providing optical access to these regions, while preserving their connections to superficial structures, 3PM enables researchers to study previously inaccessible neural circuits in their native, intact state — a crucial advantage for understanding the brain as an integrated system rather than as a set of isolated components. This advantage becomes particularly important when investigating how information flows through cortical columns, how deep and superficial circuits interact during behaviour, and how subcortical structures modulate cortical processing — questions that are at the heart of modern neuroscience and that can only be investigated using techniques that maintain the integrity of neural circuits spanning from surface to deep regions.

Although 3PM offers distinct advantages for deep brain imaging, it is important to recognize that 2PM remains the preferred choice for certain applications. For imaging at depths of up to approximately 500–600 µm in the mouse cortex, 2PM typically provides superior temporal resolution with faster frame rates and lower phototoxicity, owing to the higher repetition rates of 2PM laser sources (typically 80 MHz versus 1–2 MHz for 3PM). Studies using 3PE have, therefore, primarily focused on the cells and vasculature of the deep layers of the neocortex and white matter — regions beyond the depth limit of 2PE. The deep layers of the neocortex are accessible with 2PE when the labelling density is low, minimizing fluorescence from outside the focal plane, but experiments often require dense labelling. For example, measuring the spatial organization of the orientation tuning properties of most of the neurons in the cortex requires dense labelling with an activity-dependent fluorophore[60]. Three-photon excitation overcomes this limitation as it provides high SNR in the deep layers of the cortex even in the presence of dense labelling. Three-photon excitation has also been used to reveal, for example, differences in blood flow and vascular architecture between grey and white matter despite dense label-free THG and intravenous dye signals[61]. Furthermore, 3PM can be applied to deeper structures such as the hippocampus and striatum. Previous studies using 2PM that were focused on these brain regions had to resort to invasive approaches such as cortical aspiration and the insertion of imaging cannulae[23,62], which tend to have an adverse effect on projection fibres residing within the corpus callosum, or which remove those fibres directly[63]. Finally, 3PM might contribute to a better understanding of the functional role of deep cortical layers, such as L6, especially when combined with simultaneous imaging of upper layers.

In practice, 2PM and 3PM can be complementary rather than competitive. Recent work has demonstrated effective combined workflows in which 2PM is used to map cortical activity across broad regions with high temporal resolution, whereas 3PM is deployed selectively to investigate deeper structures, such as white matter tracts or subcortical nuclei. For example, a dual-modality approach has been used to study the migration of glioblastoma cells in the corpus callosum while simultaneously monitoring cortical cancer dynamics[57], thereby leveraging the strengths of both microscopy technologies. The choice between 2PM and 3PM should, therefore, be guided by the specific experimental questions, imaging depths required and temporal resolution needed, rather than viewing one technique as universally superior.

Neural circuit organization and function

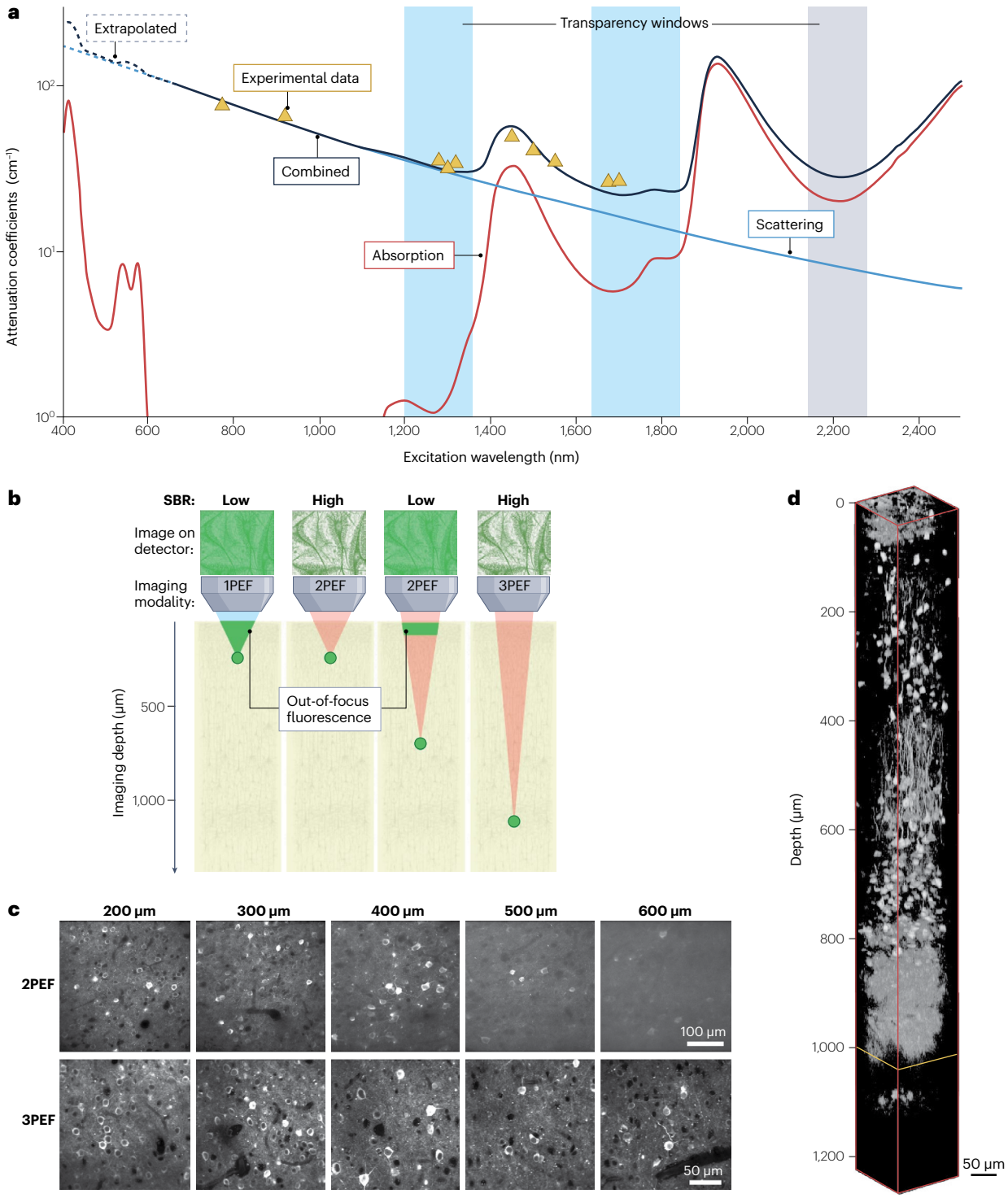
Three-photon microscopy can improve our understanding of neural circuits by enabling visualization of intact networks across cortical and subcortical depths. This capability has been particularly transformative for studying deep cortical layers and white matter, enabling the longitudinal investigation of neuronal structure and function in regions previously accessible only through invasive approaches that disrupted natural circuit organization.

Although the imaging speed of 3PM is slower than that of 2PM, it is still sufficient to record the activity of functional reporters in deep brain regions. Several studies have utilized calcium indicators to record the activity of neurons[41,54,55,64] and astrocytes[46] in deep cortical layers and in the spinal cord[47], as well as through the intact skull[41,48] of mice, highlighting the fact that 3PM is not limited to structural imaging. Although several calcium indicators, including GCaMP, rCaMP or RGECO, have been successfully used in 3PM, we note that red calcium indicators (excited by wavelengths of ~1,700 nm) have so far been reported to be less bright than green ones, although quantitative measurements of their absorption cross sections and spectra are still missing.

In mouse, in vivo 3P imaging of neurons in the subplate beneath the visual cortex has revealed unexpected insights about their role in visual processing. Despite their anatomical proximity to visual cortex layer 6, 3PM functional imaging of neurons expressing GCaMP6s showed that only 45% of subplate neurons respond to visual stimuli — markedly less than neurons in the overlying cortical column. In addition, their direction and orientation tuning were weak, further suggesting a more limited role in visual processing than their location might imply[36].

Similar insights have emerged from 3PM functional imaging studies of motor-related activity in deep cortical circuits. Layer 6 excitatory neurons in the mouse posterior parietal cortex were shown to have a distinct functional specialization during skilled movement[65]. During forepaw reaching tasks, approximately half of these neurons displayed grasp-related activity, with some increasing activity before and during the grasp whereas others activated primarily afterward. This functional organization mirrors the diversity observed in superficial layers, suggesting consistent computational principles throughout cortical depth[65].

Head-mounted 3P microscopes, when combined with genetically encoded activity markers[10], have enabled activity recordings from neuronal populations in the deep cortical layers[65,66] and upper hippocampus[65], during free, neuroethologically relevant behaviour



(Box 3). These miniaturized devices provide access to neural dynamics during unrestrained behaviour in which the animals can interact with their environment freely while having access to their full repertoire of sensory inputs, including active vision[67,68] and proprioceptive and vestibular feedback. Although the application of 3PE to head-mounted microscopes is relatively recent, their remotely operated focusing capability and their ability to be operated in both lit and dark conditions[66] mean that they can be used to quantify the spatiotemporal dynamics of neuronal subpopulations located across the cortical layers in the same animal performing the same behaviour. Thus, the link between complex behaviour and its neural circuit underpinnings can be explored.

Although most studies using 3PM in the neurosciences have so far been conducted in the rodent brain, there is increasing interest in other

Fig. 1 | Principles of three-photon microscopy for intravital imaging. a, Deep brain imaging is limited by the scattering and absorption of photons as they pass through brain tissue, which is influenced by the excitation wavelength. The graph shows the attenuation coefficient (black curve) of the mouse brain, which results from the combined effects of the water and blood absorption coefficient (red curve) and the scattering coefficient (blue curve), as a function of excitation wavelength. The scattering coefficient represented by the solid blue curve was calculated using the Mie theory of an artificial sample that mimics the mouse brain. The scattering coefficient represented by the dashed part of the blue (black) curve was extrapolated from the scattering (combined) model using an exponential fit of the data between 667 and 1,000 nm. The yellow triangles represent experimental measurements[121] of the attenuation coefficient. As shown in the graph, there are two principal transparency windows (highlighted in light blue) for three-photon excitation of fluorescence (3PEF): one at ~1,300 nm and a second at ~1,700 nm, which coincide with minimal attenuation coefficient. Another transparency window at ~2,200 nm (highlighted in grey) is rarely used as it does not excite commonly used fluorophores. **b**, The signal-to-background ratio (SBR) when imaging a fluorescently labelled column of brain tissue in different modalities. Blue-shaded and pink-shaded regions represent linear and multiphoton excitation laser light, respectively, whereas the green-shaded areas represent the part of the sample that is being excited and emitting fluorescence. The SBR is low for one-photon excitation of fluorescence (1PEF) owing to the lack of optical sectioning. Similarly, the SBR limits the imaging depth for two-photon excitation of fluorescence (2PEF) to ~600 µm owing to out-of-focus fluorescence resulting in a strong background (represented by increased green background on the detector images). The SBR is higher for 3PEF, allowing deep tissue imaging. **c**, A comparison of SBR for 2PEF and 3PEF at different imaging depths in the mouse brain. The SBR of the 2PEF images decreases with increasing imaging depth, whereas it stays high for 3PEF imaging. **d**, In vivo 3PEF deep imaging of red fluorescent protein-labelled neurons, showing that the SBR remains high beyond 1 mm of imaging depth. Part **a** adapted from ref. 121, CC BY 4.0 (https://creativecommons.org/licenses/by/4.0/); part **c** adapted from ref. 52, CC BY 4.0 (https://creativecommons.org/licenses/by/4.0/); part **d** adapted from ref. 25, Springer Nature Limited.

parts of the nervous system, such as the spinal cord[69]. Furthermore, 3PM is also starting to yield insightful data in other model organisms (Fig. 2). These include analysis of haemodynamics in brain vessels in the macaque monkey[70], volumetric imaging of sensory-related brain regions in the developing *Danionella dracula* brain[71], third harmonic generation (THG) whole-brain mapping in different fish species in vivo[72,73], large field-of-view imaging of the *D. dracula* brain[64], and the visualization of neural structure in the *Drosophila melanogaster* brain[74,75]. Functional imaging using 3PM has also been demonstrated in the highly scattering brains of the fruit fly[74,76] and zebrafish[77], and recently, label-free[78] and calcium imaging[79] in human cerebral organoids was also achieved with 3PM.

In the future, the combination of 3PM with voltage indicators optimized for multiphoton excitation, together with faster scanning strategies, promises to reveal even more detailed aspects of circuit function. When combined with multi-plane, volumetric imaging[54], these advances may soon enable monitoring of neural activity across entire cortical columns simultaneously, providing insight into how information flows through cortical layers during behaviour.

Glial biology

Three-photon microscopy has also enabled the visualization of glial cells and their dynamics in previously inaccessible brain regions, such as the white matter[80]. Previous work has focused on oligodendrocyte lineage cells in the white matter and revealed fundamental differences between the white and grey matter, highlighting higher densities of both oligodendrocytes and their progenitor cells (OPCs) and elevated OPC proliferation rates in the white matter compared to the grey matter[81,82]. Longitudinal 3PM imaging in transgene mice with fluorescent reporters for the oligodendroglial lineage has further shown that OPC generation declines more rapidly with age in white matter than in grey matter, although this region also demonstrates faster oligodendrocyte population recovery following demyelinating injury[83] (Supplementary Fig. 2).

Astrocytes have emerged as crucial regulators of brain function through their complex interactions with neurons[84,85]. These cells extend thousands of fine processes that form 'tripartite synapses' with neuronal elements, wherein they actively participate in neurotransmission through calcium-dependent release of neuroactive compounds[84]. The development of genetically encoded calcium indicators has revealed that astrocytes exhibit dynamic calcium signalling that varies significantly between brain regions and cortical layers[86]. Three-photon microscopy has enabled the application of these tools to deep-layer astrocytes, including the first recordings of calcium signalling in the fibrous astrocytes that reside in the highly scattering corpus callosum[46].

Three-photon microscopy offers unique advantages for studying astrocyte function in intact brain tissue for several key reasons. First, astrocytes show remarkable molecular and morphological heterogeneity across cortical layers and brain regions[87,88], as well as distinct calcium signalling patterns that can only be properly studied in their native environment. Second, astrocytic calcium responses are tuned by local neuronal activity[89,90], making it crucial to study these cells in the context of their natural neural circuits. Last, white matter astrocytes exhibit distinct morphologies and presumably specialized functions compared to their grey matter counterparts[91,92], necessitating imaging approaches that can access deep white matter structures without damaging nearby tissues.

Recent technical advances have expanded the toolkit for studying astrocyte function beyond calcium imaging. New optical sensors for neurotransmitters (such as iGluSnFR[93] and iGABASnFR[94]), as well as sensors of neuromodulators (such as GRAB[95]), enable simultaneous monitoring of multiple signalling molecules in and around glial cells[96]. When combined with the deep-imaging capabilities of 3PM, these tools may reveal how glial cells integrate into neural circuits across different brain regions and how they contribute to both normal brain function and disease processes.

Looking forward, 3PM could be essential for understanding how astrocytes, microglia and cells of the oligodendroglial lineage adapt to neural activity and contribute to circuit plasticity throughout life. The ability to study these cells in their native context, particularly in white matter[97] and deep cortical layers[98], may reveal new aspects of glial biology that have been inaccessible to traditional imaging approaches.

Cancer neuroscience

Three-photon microscopy has provided important insights into brain tumour biology and, in particular, into how tumours interact with deep brain structures[57,99]. This capability has been especially valuable for investigating glioblastoma. Here, invasion of the white matter by tumour cells is a hallmark of its growth pattern in the brain[100,101].

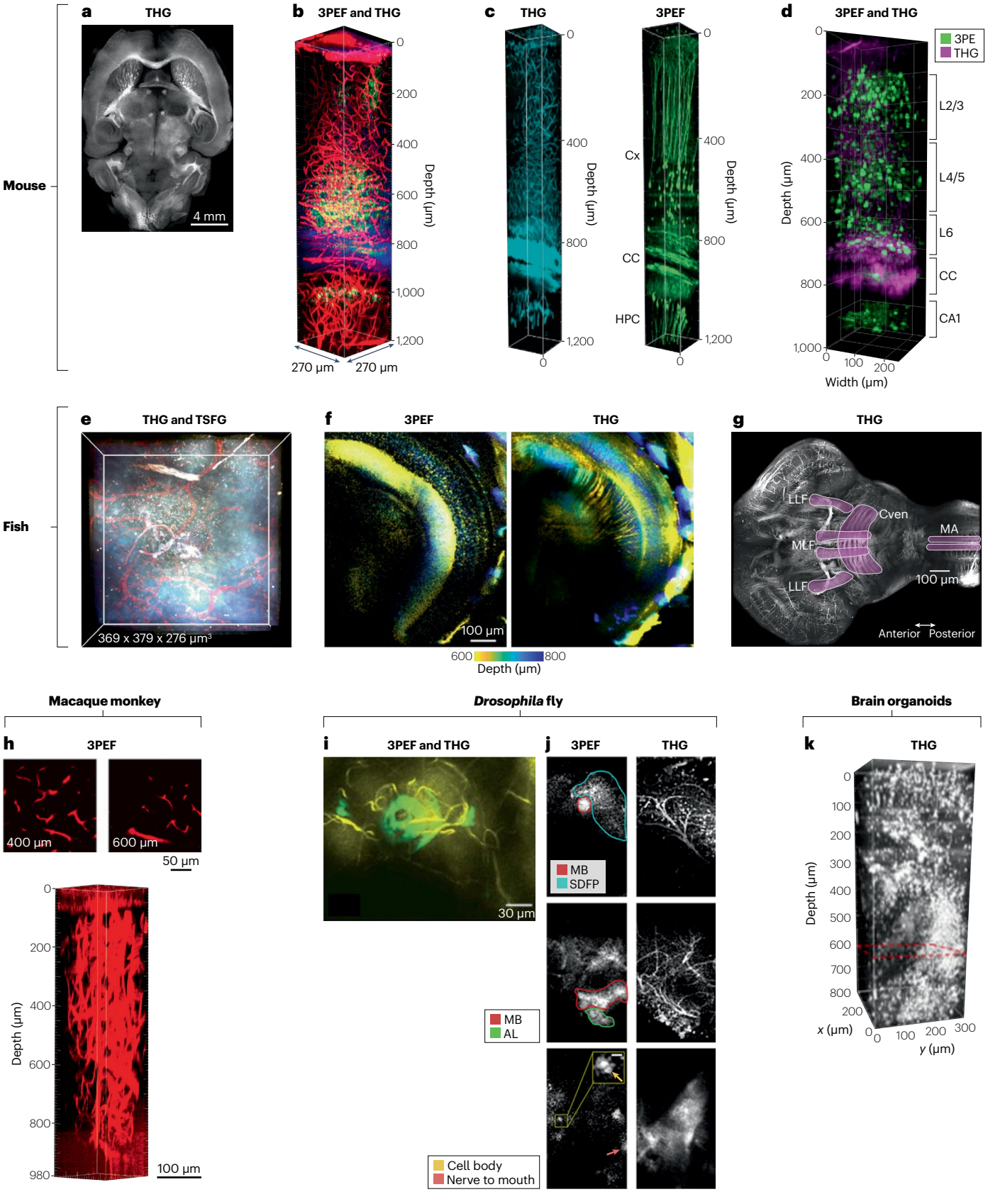

Mouse
a THG
4 mm
b 3PEF and THG
Depth (μm)
270 μm
270 μm
c THG
3PEF
Cx
CC
HPC
d 3PEF and THG
3PE
THG
L2/3
L4/5
L6
CC
CA1
Width (μm)
Fish
e THG and TSFG
369 x 379 x 276 μm³
f 3PEF
THG
100 μm
Depth (μm)
g THG
LLF
Cven
MA
MLF
LLF
100 μm
Anterior
Posterior
Macaque monkey
h 3PEF
400 μm
600 μm
50 μm
Depth (μm)
100 μm
Drosophila fly
i 3PEF and THG
30 μm
j 3PEF
THG
MB
SDFP
MB
AL
Cell body
Nerve to mouth
Brain organoids
k THG
Depth (μm)
x (μm)
y (μm)

Fig. 2 | Examples of three-photon microscopy in different animal models. **a**, Low-numerical aperture third harmonic generation (THG) imaging of a tangential ex vivo section of the mouse brain, highlighting myelinated regions. **b**, A multicolour three-photon (3P) fluorescence image of GCaMP6s-labelled neurons (green), Texas Red-labelled blood vessels (red) and THG (blue) from the mouse brain surface to a depth of 1,200 µm, highlighting the possibility of multicolour imaging with 3PM. **c**, A 3D reconstruction of a *z*-stack of images of the visual cortex (Cx), corpus callosum (CC) and hippocampus (HPC) of a mouse expressing enhanced green fluorescent protein (EGFP) in neural cells, illustrating deep 3P imaging enhanced by using 3P-adaptive optics. The left image shows the THG signal and the right image shows the EGFP signal (also see Supplementary Fig. 1). **d**, A reconstruction of a 970-µm stack of GCaMP6s-labelled neurons in layers 2/3 (L2/3), L4/5 and L6 of the posterior parietal cortex (PPC) and underlying hippocampal CA1 region. This showcases the 3PM data that can be obtained from a 3P miniscope (also see Box 3). Green shows the GCaMP6s 3P excited fluorescence; magenta shows the THG signal. **e**, Three-dimensional rendering of a z-stack of label-free third-order sum frequency generation (TSFG) images of the telencephalon of an adult zebrafish, showing the potential of TSFG to obtain multi-colour label-free contrast at depth and to discriminate different tissue components. **f**, Three-photon imaging of the optic tectum of a living, intact adult zebrafish brain, with a colourmap indicating imaging depth. **g**, Identification of major fibre bundles (the medial longitudinal fasciculus (MLF), lateral longitudinal fasciculus (LLF), ventral commissure (Cven) and Mauthner axon (MA)) and vasculature with THG microscopy in the adult brain of *Danionella dracula*. The image shows the maximum projection of all the frames that contain the MLF and highlight how THG enables label-free in vivo mapping of the whole *D. dracula* brain. **h**, Imaging of the vasculature in the deep premotor cortex of a macaque monkey. The top panels show XY images obtained at different depths. The bottom panel is a 3D reconstruction of the cortical vasculature up to a depth of ~1 mm, showing deep 3PM in a non-human primate. **i**, Image of GFP-labelled ring neurons (green) in the central complex of a *Drosophila* fly showing that 3P imaging through the cuticle is possible. Image shows a cross section at a 96-µm depth captured through the cuticle. THG imaging shows the tracheal arbors (yellow). **j**, Axially projected 3P images of the fly brain at 0–20 µm, 40–60 µm and 170–190 µm (top to bottom rows, respectively). The left column shows pan-neuronal GFP labelling of mushroom body (MB), superior dorsofrontal protocerebrum (SDFP) and antennal lobe (AL) cells, whereas the right column shows THG signal, providing clear images of the tracheal system, showing the 3P image quality obtainable in the *Drosophila* fly with the cuticle removed. **k**, Three-dimensional reconstruction of THG imaging of a live human cerebral organoid showing the potential of 3PM to allow cell tracking in a label-free manner in these new models. Part **a** adapted with permission from ref. 114, Elsevier; part **b** adapted with permission from ref. 55, AAAS; part **c** adapted from ref. 46, Springer Nature Limited; part **d** adapted from ref. 65, Springer Nature Limited; part **e** adapted from ref. 73, Springer Nature Limited; part **f** adapted from ref. 64, Springer Nature Limited; part **g** reprinted with permission from ref. 72, Wiley; part **h** adapted with permission from ref. 70, Elsevier; part **i** adapted from ref. 74, CC BY 4.0 (https://creativecommons.org/licenses/by/4.0/); part **j** adapted with permission from ref. 75, Optica; part **k** adapted from ref. 78, CC BY 4.0 (https://creativecommons.org/licenses/by/4.0/).

Direct visualization of tumour cell dynamics in the corpus callosum via adaptive optics-enhanced 3PM has revealed that there are distinct invasion properties in this region, as compared to cortical regions. Although glioblastoma cells in both brain regions can form multicellular tumour networks via gap junction-coupled or adherens junction-coupled contacts with other tumour cells, their invasion patterns show marked differences[57] (Supplementary Fig. 2). In the corpus callosum, cells exhibit more directed migration along myelinated fibres. By contrast, in the cortex, more glioblastoma cells invade via a process in which their neurite-like processes branch out. Interestingly, the tumour–tumour-coupled networks observed in the corpus callosum align with white matter architecture, demonstrating adaptation to the local microenvironment[57].

More recently, a preprint paper has described the use of 3PM to illuminate how the immune system responds to tumour infiltration in deep brain structures[99]. According to this report, microglial responses in the corpus callosum show enhanced surveillance in regions of sparse tumour infiltration but reduced activity in densely infiltrated areas, suggesting potential region-specific immune responses to primary brain tumours.

The ability of intravital multiphoton imaging to maintain resolution at depth has proven particularly valuable for studying neuron–tumour interactions[102–104]. In the future, neuronal hyperexcitability, a phenomenon that often occurs during glioma progression, could be monitored in deep cortical layers and subcortical regions such as the hippocampus with 3PM and calcium or voltage reporters. This will potentially provide new insights into how tumours influence neural circuit function and contribute to seizure activity.

From a clinical-translational perspective, 3PM findings have highlighted the limitations of current radiological imaging approaches. Early tumour infiltration of white matter tracts often avoids detection by conventional magnetic resonance imaging (MRI)[105–107], suggesting the need for more sensitive diagnostic tools. The detailed cellular insights from 3PM may help identify new imaging biomarkers for early detection of white matter invasion.

Looking forward, 3PM promises to advance our understanding of treatment resistance and therapy-induced changes in the tumour microenvironment. The ability to monitor cellular responses to therapy in deep brain regions may reveal new mechanisms of treatment failure and suggest novel therapeutic strategies.

Neuroimmune interactions and disease

Three-photon microscopy may also open new avenues for studying immune cell dynamics in previously inaccessible brain compartments[98,99]. This capability may prove particularly valuable for investigating immune responses across multiple neural interfaces, including the bone marrow, meninges, choroid plexus and white matter tracts.

Deep 2PM has revealed previously unknown aspects of immune cell trafficking in neuroinflammatory conditions[69,108,109]. In the future, using 3PM, researchers will be able to simultaneously visualize immune cell recruitment, migration and interaction with neural tissue in intact preparations. This approach has already been informative in models of multiple sclerosis, in which 3PM enabled direct observation of demyelination and remyelination processes in both cortical and subcortical regions while maintaining tissue integrity[83].

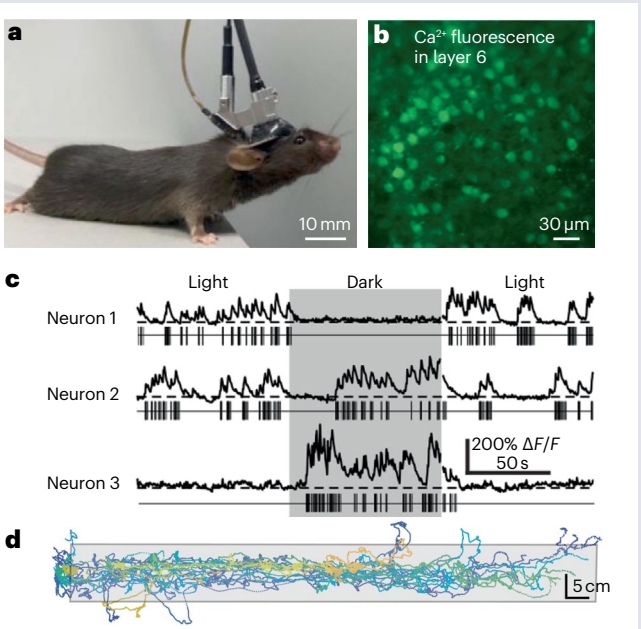
The choroid plexus–cerebrospinal fluid interface has emerged as a crucial checkpoint for immune cell entry into the CNS. Invasive 2PM has illuminated the sequential steps of myeloid cell recruitment and patrolling at this barrier, providing insights into how immune responses are initiated and regulated in the brain[110]. In turn, 3PM has revealed immune cell dynamics in lymphoid tissues, allowing visualization of CD4 and CD8 T cell behaviours throughout entire unperturbed lymph nodes at depths of 600–900 µm (ref. 111) (Supplementary Fig. 2). Thus, deep 3PM could enable a non-invasive approach to examine strategically important regions of the immune compartment in the nervous system.

Box 3 | Head-mounted three-photon microscopes

Miniature microscopes[164], light enough to be mounted on the heads of rodents without any additional effects on their behaviour (see the figure, part **a**), circumvent the constraints of the head fixation that is traditionally required for in vivo microscopy and enable the imaging of large populations of neurons in freely behaving animals.

Unlike single-photon excitation-based wide-field imaging miniscopes[165,166], head-mounted microscopes using two-photon excitation (2PE)[167] enable optical sectioning within the upper cortical layers[168,169]. When combined with fluorescent activity reporters[10], they also allow measurement of neuronal circuit activity while animals interact with their environment in a self-determined way[170,171]. However, owing to the smaller optics in miniature head-mounted microscopes and scattering within the tissue, the practical depth limit for 2PE head-mounted microscopes is around 400–500 µm (ref. 169), shallower than the theoretical[29] and functional limit for conventional microscopes[172]. More recently, three-photon excitation (3PE) has been applied to head-mounted microscopes to gain access to deep cortical layers in freely moving rats[173] and mice[65,66] (see the figure, part **b**) and superficial layers of the hippocampus[65].

Like all head-mounted multiphoton microscopes[167–169], the laser-sourced excitation light for 3PE is delivered to the microscope through a glass fibre. To enable laser pulses with a temporal profile of ~50 fs to be delivered into the tissue, the beam passes through silicon prisms and thick silicon slabs to compensate for second-order and third-order dispersion[173]. Lastly, given the inherent fibre-motion and twisting brought about by freely moving animals, the transmission properties of the delivery fibre must be motion-insensitive and able to retain polarization; this is achieved using hollow-core fibres[174]. Additional developments, such as on-board photon detection[66], have further reduced the tethering burden to the animal. In particular, silicon photon-multipliers can be mounted directly onto the head-mount. Their fast response allows gating for optical noise rejection, significantly improving the SNR[66] and potentially also enabling multiplexed readout of multiple imaging channels[175]. To further reduce the need to interfere with the animal, remote lightweight focusing mechanisms have been added to 3PE microscopes[65,66], enabling 3D images to be collected throughout the cortex while mounted. These remotely operated mechanisms can either change the image plane during behaviour[66] or image activity



from multiple planes[65] (the latter based on an approach first used in a 2PE head-mounted microscope[176]). Using these approaches, head-mounted 3PE microscopes enable long-term recording from hundreds of cortical neurons over multiple days, and they can sample from neuronal populations across the cortical depth during a wide range of neuroethologically relevant behaviours. Part **c** of the figure shows neural recordings during a light on ('Light') and off ('Dark', grey-shaded region) paradigm, whereas part **d** shows the behavioural tracking during the imaging session (where colour encodes time). Future developments will leverage recent progress in conventional multiphoton microscopes such as multiplexing and simultaneous imaging of multiple planes or fields of view[175]. Advances may include the addition of behavioural tracking cameras and the combination of 2PE and 3PE[54] with cellular resolution photostimulation[177]. Part **a** is adapted from ref. 65, Springer Nature Limited. Parts **b**–**d** are adapted from ref. 66, Springer Nature Limited.

Beyond inflammatory conditions, 3PM can advance our understanding of immune responses in neurodegenerative diseases[107,112]. The ability of the technique to image deep brain structures can reveal how immune cells interact with pathological protein aggregates and respond to neuronal damage in regions previously inaccessible to in vivo imaging. This can provide new insights into disease progression and may lead to potential therapeutic interventions.

In addition to imaging of immune cells, blood vessel dynamics[113] and myelinated tract integrity[114] can be monitored simultaneously using THG imaging, enabling correlations between vascular changes, immune cell recruitment, and tissue pathology to be identified.

Looking forward, the integration of 3PM with new molecular probes and faster imaging protocols promises to further expand our understanding of neuroimmune interactions. These advances may soon enable real-time tracking of immune cell activation states and their influence on neural circuit function in both healthy and diseased conditions.

Future technical advances and directions

Recent technical developments in 3PM have substantially expanded its capabilities for neuroscience applications. Adaptive optics has proven particularly transformative, enabling imaging at depths beyond 1.4 mm in the mouse brain with near-diffraction-limited resolution[46,47,115–117] (Supplementary Fig. 1). Although adaptive optics correction so far is slow and indirect, future work exploiting brighter, 3P-tailored infrared guidestars[56,118] and direct wavefront sensing techniques[119] may further improve correction quality, speed and achievable depth. In the future, optical wavefront shaping[120,121]

has the potential to advance deep-imaging possibilities beyond the realm of adaptive optics: the first proof-of-concept work has now extended wavefront shaping from 1PE and 2PE regimes to the 3PE regime[122]. These technical advances increase the capabilities of 3PM and opportunities for neuroscientists; however, it is important to realize that implementing them may come with varying practical challenges and often considerable costs. For this reason, many are often not readily available outside the laboratories of developers (although the first commercial, advanced 3P systems are starting to appear). Thus, the complexity, cost and effort need to be carefully considered when selecting an imaging modality, as well as whether advanced 3P system performances are necessary to answer the biological question at hand.

Looking forward, several emerging capabilities promise to further expand 3PM applications. The combination of higher-repetition-rate lasers with fast-scanning mechanisms, such as acousto-optic modulators[123,124], may soon enable voltage imaging of targeted and/or selected neurons in deep brain structures in a random-access fashion, providing direct readouts of neural activity with millisecond precision. To increase the repetition rate of 3P lasers, further innovations in femtosecond laser technologies are, however, required. These could involve new laser substrate types, improvements in supercontinuum light generation efficiency, and advances in non-collinear optical parametric amplifier systems. Three-photon microscopy of voltage indicators, although so far not experimentally demonstrated, is in principle possible despite the challenging requirements for speed and sampling (~kHz) to resolve the highly transient voltage signal (~ms). For example, 1-MHz pulses would give access to ~1,000 individual points, that is, voxels, that could be sampled at 1 kHz each, which shows the in-principle feasibility of using voltage indicators in the future. Moreover, the development of dedicated large field-of-view systems[64,125] with adequate optical design and objectives[126] could allow simultaneous imaging across multiple cortical columns and brain regions at depth, similar to 2P mesoscopes[127–131].

The integration of 3PM with other technologies presents additional opportunities (Fig. 3). Combining 3PM with clinical imaging modalities such as MRI could help identify new biomarkers for early and improved disease detection. The integration of 3PM with emerging molecular tools and sensors, such as calcium and voltage indicators and emerging genetically encoded sensors[93,94,132], that are compatible with intravital imaging[21,124] offers potentially powerful approaches to reveal new aspects of cellular function and communication in deep brain structures. The list of available fluorophores and reporters that can be efficiently excited by 3P is bound to increase, and the generally broad 3P excitation and broad laser light spectra facilitate the combination and simultaneous excitation of fluorophores, indicators and actuators[55,133] (Supplementary Table 1). At the same time, the broad 3P excitability of fluorophores in the near-infrared region of the spectrum can be disadvantageous, as different multiphoton processes can be simultaneously induced with a single laser line[31]. Thus, judicious choices of dichroic mirrors and filters are called for, as well as unmixing of fluorophores during (real-time) imaging processing. Here, recent developments in AI-based approaches to unambiguously discriminate different structures giving raise to the same THG signal[57] and for more

Glossary

Absorption cross section
A measure of the probability that a photon will be absorbed to excite a (fluorescence) molecule, determining the efficiency of fluorescence signal generation.

Adaptive optics
A technology that corrects for wavefront distortions caused by the sample or imaging system, thereby improving image resolution and signal in deep and/or scattering biological tissues.

Attenuation coefficient
A parameter describing the reduction in light intensity as it propagates through a medium, owing to scattering and absorption.

Confocal microscopy
An imaging technique that uses a spatial pinhole to block out-of-focus light, thereby producing high-contrast optical sections of thick specimens.

Dual-plane imaging
A microscopy method that simultaneously captures images from two different focal planes, enabling faster acquisition of three-dimensional data.

Guidestars
Artificial or natural reference points emitting (fluorescence) signals, used in adaptive optics systems to measure and correct wavefront distortions for improved resolution and image quality.

Multiphoton microscopy
An advanced imaging method wherein two or more low-energy photons are absorbed simultaneously to excite a fluorophore, enabling deep tissue imaging with high resolution and minimal photodamage.

Optical sectioning
The ability of an imaging system to isolate thin slices of a sample along the axial direction, improving resolution and contrast in three-dimensional imaging.

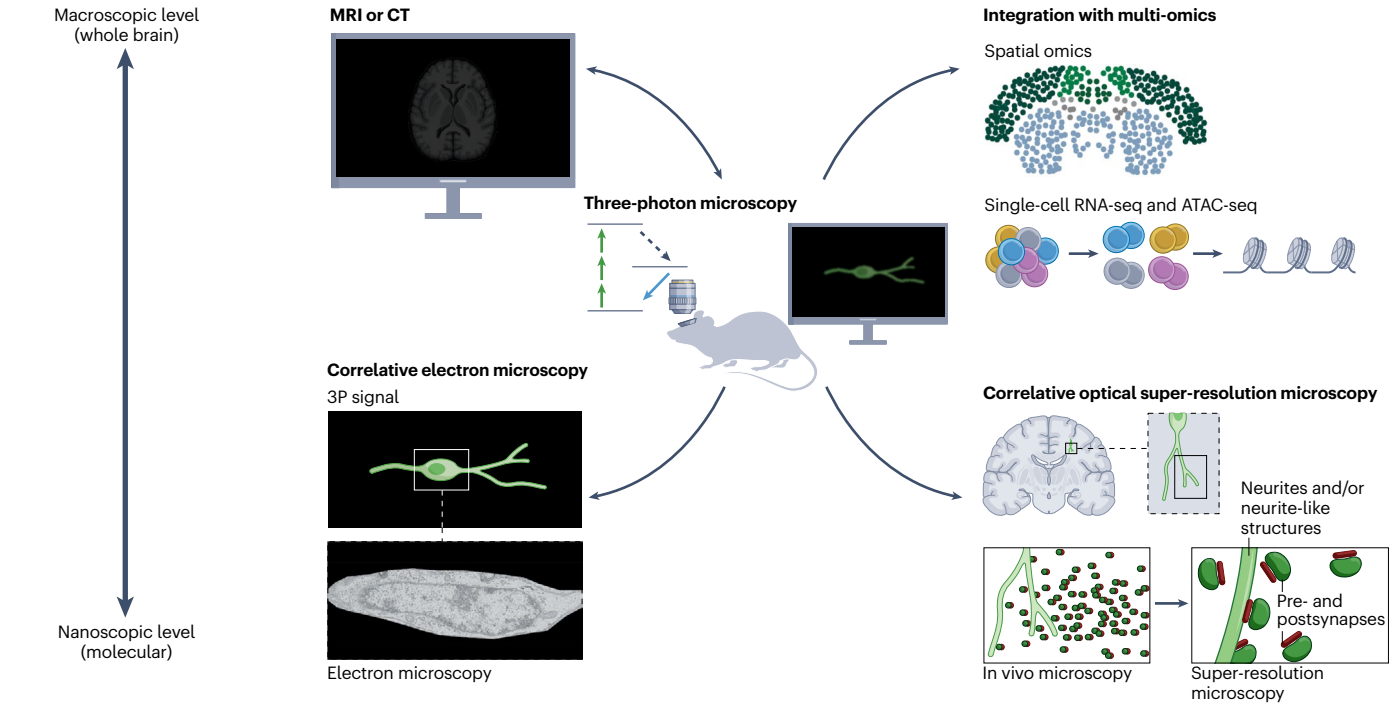
Optical wavefront shaping
The manipulation of light wavefronts using devices such as spatial light modulators to control and correct for aberrations introduced by complex, highly scattering media.

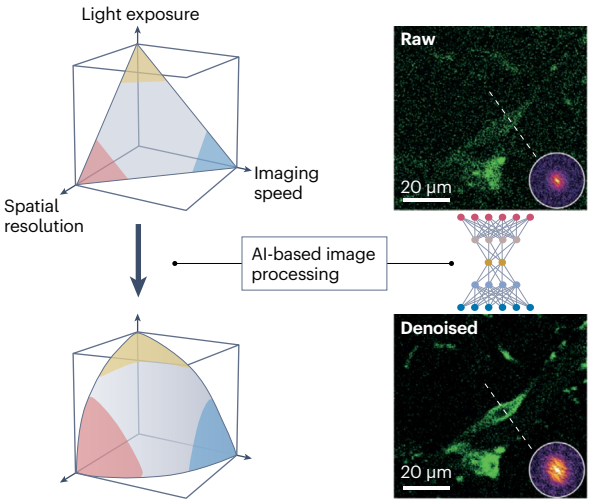
Remote focusing
A technique to adjust the focal plane of a microscope rapidly and precisely by manipulating the optical path without moving the primary objective lens.

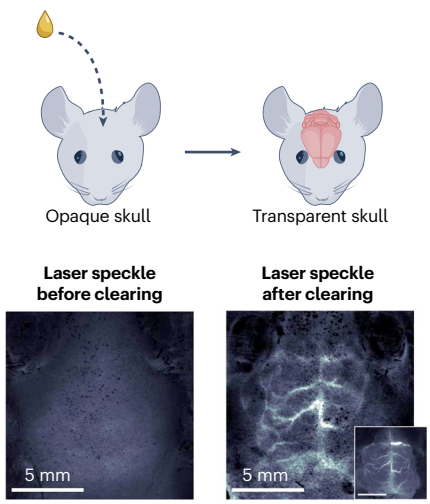
Third-order harmonic signal
A nonlinear optical signal generated when three photons combine to produce a single photon with triple the energy (or one-third the wavelength), enabling visualization of intrinsic sample structures without fluorescent labels.

Volumetric imaging
A method for capturing 3D data sets by acquiring images across multiple depths, enabling detailed reconstruction of structures in three dimensions.

Wavefront distortions
Aberrations in the light wavefront caused by refractive index variations in a sample, degrading image quality in microscopy.







general decomposition[134] of overlapping fluorescence signals may be crucial for multi-colour imaging applications.

In the near future, the combination of optogenetic stimulation with 3PM could enable researchers to measure and manipulate populations of cells deep within the brain, testing causal links between cellular activity and animal behaviours[135]. For this, a multi-laser system for simultaneous two-photon and three-photon excitation could leverage the strengths of both techniques: for example, 2PE could be used to stimulate in upper cortex, whereas the output is recorded in deeper brain regions via 3PM, thereby substantially enhancing the capabilities of all-optical electrophysiology.

Another transformative avenue in neuroscience pertains to the linking of neural dynamics to their underlying molecular ultrastructure. This necessitates correlative intravital imaging methods and workflows that are capable of bridging the involved spatial scales, from nanometres to millimetres. Recent advances in expansion microscopy[136] and its combination with indiscriminate 'pan'-labelling for protein density[137,138] have allowed 3D cellular architecture to be reconstructed at nanoscale and synapse-level resolution[139]. Similarly, 2PM-based intravital neural recordings have been combined with synchrotron X-ray holographic nano-tomography of the same brain tissue that provides context over millimetre-scale volumes with sub-100-nm resolution. This was followed by targeted acquisition of serial block-face electron microscopy volumes to acquire the ultrastructure of the tissue[140,141]. We reason that the introduction of 3PM into these cross-scale workflows could readily expand the repertoire of possible experiments, by providing

Fig. 3 | Future directions for three-photon microscopy. a, Correlative imaging approaches and the integration of three-photon microscopy (3PM) with multi-omics approaches will allow researchers to bridge images across scales, from the nanoscale to the macroscale. Macroscopic clinical imaging techniques such as magnetic resonance imaging (MRI) and computed tomography (CT) and/or nanoscopic methods such as electron and super-resolution microscopy can be used for correlative imaging, enabling the same regions and cells to be imaged across scales in space and time. Furthermore, technologies that will allow imaging data to overlay the molecular information gained via spatial and/or single-cell RNA-sequencing or proteomics will be an important future direction. **b**, In 3PM, key trade-offs between light exposure, spatial resolution and imaging speed have to be made to respect sample health. These are depicted as opposing corners of a pyramid in the upper left panel. Artificial intelligence (AI)-based image processing can be used to restore aspects of the image that are lost owing to these trade-offs and, thus, expands the possibilities for optimizing these trade-offs (illustrated as enlarged corners of the pyramid in the lower left panel). The images on the right show a glioblastoma cell imaged via 3PM. The upper image shows the raw 3PM image of the cell, and the lower image shows the same image after U-Net-based denoising. Insets show spectral power maps. **c**, In vivo clearing of live tissue increases imaging depth. By applying aqueous solutions of dye molecules that exhibit strong absorption of excitation wavelengths in the blue and near-ultraviolet regions to the tissue, optical transparency can be improved in the red spectrum. The images at the bottom illustrate laser speckle contrast, an imaging technique used to visualize dynamic processes, such as erythrocyte flow within blood vessels, before (left) and after (right) the topical application of dye, demonstrating an effective increase in transparency and imaging of blood vessels with an intact skull. The inset at the bottom right shows the laser speckle contrast after removal of the skull for comparison. Electron microscopy image in part **a** is adapted with permission from ref. 103, Elsevier. Brain imaging and brain slice schematics in part **b** were created in BioRender, Venkataramani, V. (2025). Left images in part **b** are adapted from ref. 142, Springer Nature Limited. Right images in part **b** are adapted with permission from ref. 57, Springer Nature Limited. Part **c** is adapted with permission from ref. 148, AAAS.

molecular and structural context to dynamic intravital experimental paradigms (Fig. 3a).

Computational advances, particularly related to AI-based image processing, are poised to further enhance 3PM capabilities. Deep learning-based denoising approaches that are specifically optimized for the noise characteristics of 3PM data have demonstrated remarkable improvements in image quality, effectively increasing imaging depth by computationally recovering signals that would otherwise be lost in noise. For example, recent U-Net-based architectures trained on paired noisy and high-SNR 3PM images have the ability to restore fine structural details from degraded deep tissue images, enabling visualization of small cellular processes that were previously indiscernible at comparable depths[142,143]. Similarly, computational adaptive optics[144] that use the image data itself to estimate and correct for aberrations have shown promise for enhancing 3PM resolution without requiring complex hardware modifications (Fig. 3b). These approaches are particularly valuable for 3PM, wherein traditional wavefront sensing becomes challenging at greater depths owing to reduced photon budgets. Beyond image enhancement, AI-driven 'smart microscopy' approaches that adaptively modify acquisition parameters based on real-time analysis of the incoming data are beginning to transform how imaging experiments are conducted. These systems can automatically identify regions of interest, detect rare cellular events, and optimize imaging parameters to minimize photodamage while maximizing information content[145–147]. For 3PM specifically, such intelligent acquisition strategies are crucial for balancing the competing demands of signal strength, imaging speed and tissue viability when operating at the edge of what is physically possible in terms of imaging depth. The integration of these computational approaches with 3PM hardware advances creates a synergistic effect, in which the software can effectively extend the practical capabilities of the hardware beyond what is possible through optical engineering alone. This may make previously impossible neuroscience experiments feasible and expand the range of questions that can be addressed with this technology.

One of the most profound limitations to achievable imaging depth and resolution is the scattering that arises from tissue heterogeneity. To overcome this, novel approaches to achieving optical transparency in live animals have been put forward, with demonstrations of both cranial and extra-cranial application[148–150] (Fig. 3c). When paired with 3PM, this could effectively enhance practical imaging depth and resolution and enable optical access to deep cortical and subcortical areas through the intact skull[151] or deep-seated, subcutaneous organs, which will be transformative for studies of extra-cranial nerves and tumours.

Lastly, although 3PM has been applied across many model organisms, its translation to human neuroscience represents both an exciting frontier and a considerable challenge. Direct monitoring of healthy neural circuits in humans using 3PM remains a distant prospect, constrained by substantial technical, safety and ethical considerations. More immediate translational opportunities exist in clinical neurosurgery. The recent implementation of confocal laser endomicroscopy in neurosurgical settings[152] provides a precedent for advanced optical technologies in the operating room. In this context, the label-free THG capabilities of 3PM could offer distinct advantages by enabling real-time identification of axonal fibre tracts and healthy neuronal tissue without requiring exogenous contrast agents. Such applications could significantly improve the precision of tumour resections while preserving critical neural structures, potentially reducing postoperative deficits. In the context of brain surgery, various in vivo and ex vivo microscopy technologies have been successfully used as guidance tools[153–155]. We anticipate that similar applications with 3PM could be another frontier for technology development. Although full-scale 3PM integration into clinical practice requires substantial technological adaptation and miniaturization, these targeted applications represent a more feasible near-term path for translating 3PM technology to human neuroscience with meaningful clinical impact.

Conclusion

In recent decades, 3PM has emerged as a promising key technique for deep intravital brain imaging, providing unprecedented access to deep cortical and subcortical structures while maintaining high spatial resolution at the subcellular level. Despite its potential, 3PM still faces several limitations that warrant attention. Imaging speed remains a bottleneck owing to the low repetition rates required for efficient 3PE, and meticulous parameter optimization is necessary to prevent phototoxicity and heating and to preserve tissue integrity during deep imaging. However, technical innovations, such as improved laser designs, adaptive optics and computational enhancements, are rapidly addressing these challenges. Although the need for specialized and costly equipment, such as high-power femtosecond lasers and adaptive optics, continues to limit widespread adoption, the introduction of the first commercial

systems and their availability in advanced imaging facilities mark a pivotal moment for neuroscience research. Emerging applications, from mapping neural circuits to studying glial dynamics and tumour microenvironments, underscore the potentially broad impact of 3PM across disciplines. Here, the ability of the technique to probe cellular and molecular processes in their native context, non-invasively and deep inside the brain, can open new avenues for studying the complexities of the brain and its interactions with systemic diseases.

Looking forward, the integration of 3PM with other imaging modalities, molecular probes and AI holds great potential to further push the frontiers of neuroscience across scales. As these developments unfold, 3PM might become a key tool for uncovering the mysteries of the brain and advancing both fundamental research and clinical applications.

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Author contributions

R.P., J.F.O., J.W., S.J.S. and V.V. researched data for the article. R.P., J.N.D.K., J.W., M.M. and V.V. contributed substantially to the discussion of the content. R.P., J.F.O., J.N.D.K., J.W., M.O.B., B.D., S.J.S. and V.V. wrote the article and reviewed and/or edited the manuscript before submission. R.P. and V.V. coordinated and led the writing of the article.

Competing interests

The authors declare no competing interests.

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